# ARL Spec: On-demand Power Generation

Brendan Griffiths

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# 1 Service overview

The On-demand Power Generation (OPG) service is the ARL's power plant: a solar array, a battery bank and diesel backup generators that together supply the whole laboratory. Topic prefix: `arl/opg`.

| Entity | Kind | Plant |
|---|---|---|
| `global` | `service` | The plant as a whole: total generation, total load, supply mode, generation priority, load shedding. |
| `solar` | `array` | 1,200 kWp solar array with a central inverter (up to 1,000 kW AC), south of the lab. |
| `batteries` | `bank` | 4,000 kWh lithium battery bank, up to 800 kW charge or discharge. |
| `diesel` | `generators` | Two 250 kW diesel generators, reported together as one entity, with a 20,000 L fuel tank. |

## 1.1 General state and routines

- **Reporting**: every entity reports once a minute (60 s) on its `power` stream.
- **Load**: the lab's demand follows the time of day and the weather (heating and cooling), the activity of the other laboratory services, and load shedding.
- **Generation priority**: each minute the plant covers the load from its sources in the order set by the generation priority (`solar_first` by default). Surplus solar charges the batteries; anything beyond that is curtailed. The batteries are not discharged below the battery reserve except in an emergency.
- **Automatic backup**: the generators start automatically when the batteries reach the reserve and the load is not covered, and stop again once the batteries have recovered and solar covers the load. A manual start or stop holds until the next automatic trigger.
- **Generators**: a start takes 30 s (`starting`) and can fail. A stop runs the generators unloaded for a 60 s cool-down. They burn fuel in proportion to their output.
- **Refuelling**: a fuel delivery of up to 15,000 L arrives every 7 days at 10:00 local (capped at the tank size), published as a `refuel` event.
- **Solar**: output follows the sun and the weather, and falls slowly as dust builds up on the panels; rain cleans them.
- **Load**: 4 messages a minute (about 0.07 messages/s), plus state changes, refuels and alerts.

## 1.2 Message conventions

Every service follows the same MQTT layout. `<entity>` is an entity id (or `global` for the whole service).

| Topic | Retained | Content |
|---|---|---|
| `arl/opg/$topology` | yes | Machine-readable list of the entities, streams (with field names, types, units and periods), state items, events and controls of the service. Re-published when it changes. |
| `arl/opg/<entity>/telemetry/<stream>` | no | Periodic measurements. |

| Topic | Retained | Content |
|---|---|---|
| `arl/opg/<entity>/state/<item>` | yes | The current value of a state item, re-published when it changes. |
| `arl/opg/events/<kind>` | no | One-off events, including `alert`. |
| `arl/opg/cmd/<entity>/<control>` | no | Commands sent by the platform. |
| `arl/opg/ack/<command_id>` | no | Acknowledgements of a command. |

Telemetry and event payloads carry an envelope: `v` (schema version, integer, currently 1), `ts` (simulated time, Unix milliseconds), `seq` (per-topic sequence number, restarts with the service; gaps reveal lost messages) and `entity` (entity id). The measurement fields are added next to the envelope. A measurement that cannot be taken is `null`. State payloads are `{"v", "ts", "entity", "item", "value"}`.

**Commands** are JSON `{"command_id": <string>, "researcher": <researcher code>, "value": <value>}`. The researcher must be verified by the ARL API for the service. Acknowledgements carry `command_id`, `target`, `control`, `researcher`, `state` (`accepted`, `rejected`, `in_progress`, `completed`, `failed`), `progress` (0 to 1 for long-running controls), `reason` and `detail`. Rejection and failure reasons: `controls_disabled`, `malformed`, `unknown_control`, `unknown_target`, `unknown_researcher`, `researcher_defunct`, `researcher_not_permitted`, `api_unavailable`, `invalid_value`, `precondition_failed`, `busy` (the target is already running a long command), `fault`, `timeout`, `interrupted`.

**Alerts** are events on `arl/opg/events/alert` with `alert_id`, `type` (the alert key below), `severity` (`info`, `warning`, `critical`), `state` (`raised` or `cleared`), `value`, `limit`, `unit`, `message` and `metadata`. An alert that escalates is raised again with the higher severity under the same `alert_id`. Limits come from the ARL API (`/v1/alert-limits/` + `opg`).

Every service also publishes these:

| Measurement | Key | Level | Type / unit | Description |
|---|---|---|---|---|
| Online | `online` | Global | boolean | Retained state. `true` while the service runs, `false` when it stops or loses its broker connection (last will). |
| Active alerts | `active_alerts` | Global | array of alert ids | Retained state. The ids of the alerts currently raised, in the order they were raised. |

# 2 Transmitted data

## 2.1 Telemetry (`arl/opg/<entity>/telemetry/power`, every 60 s)

| Measurement | Key | Level | Type / unit | Description |
|---|---|---|---|---|
| Total generation | `total_generation_w` | Global | integer, watts | Power produced by all sources together. |
| Total load | `total_load_w` | Global | integer, watts | Power demanded by the laboratory. |

| Measurement | Key | Level | Type / unit | Description |
|---|---|---|---|---|
| Supply mode | `supply_mode` | Global | enum | `solar`, `battery` or `diesel` when one source supplies more than 90% of the load, else `mixed`. |
| Solar output | `power_output_w` | Entity `solar` | integer, watts | AC power delivered by the solar array. |
| Solar irradiance | `irradiance_wm2` | Entity `solar` | integer, W/m² | Solar radiation on the array. |
| Panel temperature | `panel_temperature_k` | Entity `solar` | float, kelvin | Temperature of the panels. Hot panels produce a little less power. |
| Inverter status | `inverter_status` | Entity `solar` | enum | `online`, `offline` (array disabled) or `fault`. |
| State of charge | `state_of_charge_pct` | Entity `batteries` | float, % | Energy stored as a percentage of capacity. |
| Battery power | `battery_power_w` | Entity `batteries` | integer, watts | Power flowing into (positive, charging) or out of (negative, discharging) the bank. |
| Battery temperature | `battery_temperature_k` | Entity `batteries` | float, kelvin | Temperature of the battery bank. Rises with heavy charging or discharging. |
| Battery health | `battery_health_pct` | Entity `batteries` | float, % | Remaining capacity relative to new. Falls slowly with use. |
| Fuel level | `fuel_level_pct` | Entity `diesel` | float, % | Diesel in the tank as a percentage of 20,000 L. |
| Generator output | `power_output_w` | Entity `diesel` | integer, watts | Power delivered by the generators. |
| Generator status | `generator_status` | Entity `diesel` | enum | `off`, `starting`, `running` or `fault`. |
| Generator runtime | `runtime_h` | Entity `diesel` | float, hours | Total running hours of the generators. |
| Heartbeat | `ts` (envelope) | Each entity | timestamp, ms | Time of the entity's latest message. |

## 2.2 State (retained)

| Measurement | Key | Level | Type / unit | Description |
|---|---|---|---|---|
| Generation priority | `generation_priority` | Global | enum | `solar_first`, `battery_first` or `diesel_first`. |
| Load shedding | `load_shed` | Global | boolean | Whether non-critical load is being shed. |
| Solar enabled | `enabled` | Entity `solar` | boolean | Whether the solar array is enabled. |
| Battery reserve | `reserve_pct` | Entity `batteries` | float, % | Minimum state of charge kept for emergencies. |

## 2.3 Events

| Measurement | Key | Level | Type / unit | Description |
|---|---|---|---|---|
| Refuel | `arl/opg/events/refuel`: `litres`, `fuel_level_pct` | Entity `diesel` | integer litres, float % | A fuel delivery arrived: litres added and the fuel level afterwards. |
| Alert | `arl/opg/events/alert` | Global / entity | alert object | Alert raised or cleared (see *Alerts*). |

# 3 Controls

| Control | Key | Level | Value (type / unit) | Description |
|---|---|---|---|---|
| Set Generation Priority | `generation_priority` | Global | enum: `solar_first`, `battery_first`, `diesel_first` | Order in which the sources cover the load. `solar_first` (default) uses solar, then batteries, then diesel; `battery_first` draws on the batteries before solar (to cycle the bank); `diesel_first` loads the running generators first (for generator testing). Applies from the next minute. |
| Load Shedding | `load_shed` | Global | boolean | Sheds non-critical load (part of the housing and lab load and the water treatment plant). |
| Enable Solar Array | `enable_solar_array` | Entity `solar` | boolean | Enables or disables the array. Disabled, the inverter is `offline` and produces nothing. |
| Set Battery Reserve | `battery_reserve` | Entity `batteries` | float, %, 10 to 80 | Minimum state of charge kept for emergencies. |
| Start Generator | `start_generator` | Entity `diesel` | boolean, must be `true` | Starts the generators: `in_progress` while starting (30 s), `completed` when running, `failed` (`start_failed`) if the start fails. Rejected if fuel is below 2% (`low_fuel`) or they are already running. |
| Stop Generator | `stop_generator` | Entity `diesel` | boolean, must be `true` | Stops the generators after a 60 s cool-down. Only while running. Accepted even if it leaves the load uncovered. |

# 4 Alerts

Every alert carries the current `supply_mode` in its `metadata`.

| Alert | Key | Level | Severity, limit key (default, unit) | Description |
|---|---|---|---|---|
| Supply Shortfall | `supply_shortfall` | Global | critical; no limit | The load cannot be covered by the available generation. `metadata.unserved_w` is the uncovered load (W). Clears after 5 minutes covered. |
| Backup Active | `backup_active` | Global | info; no limit | The diesel generators are supplying power. Clears when they are off. |
| Solar Underperformance | `solar_underperformance` | Entity `solar` | warning; `opg.solar.min_ratio` (0.8, ratio) | With irradiance above 200 W/m², the array's output is below the limit ratio of its expected output for 15 min. `metadata.expected_w` and `metadata.output_w`. Clears above the ratio plus 0.05 for 15 min. |
| Low Battery | `low_battery` | Entity `batteries` | warning, critical; `opg.battery.warn_pct` (30 %), critical below the battery reserve | State of charge below the warning limit; critical when below the reserve (`metadata.reserve_pct`). Clears 5 points above the warning limit. |

| Alert | Key | Level | Severity, limit key (default, unit) | Description |
|---|---|---|---|---|
| Battery Temperature | `battery_temperature` | Entity `batteries` | warning; `opg.battery.max_temp_k` (318 K) | Battery temperature above the limit for 10 min. Clears 3 K below it. |
| Low Fuel | `low_fuel` | Entity `diesel` | warning; `opg.fuel.min_pct` (25 %) | Fuel level below the limit. Clears when refuelled above it. |
| Generator Fault | `generator_fault` | Entity `diesel` | critical; no limit | A generator start failed or a running generator faulted. Clears on the next successful start. |

# 5 Configuration

The service reads a TOML file (`config/` + `opg.toml` in the bundle, `--config` or `ARL_CONFIG` for a binary). Environment variables override the file, and command-line options override both (see *Running the ARL*). Defaults are the values in the provided file.

### 5.0.1 Common keys

| Key | Type / unit | Default | Description |
|---|---|---|---|
| `seed` | integer | `1` | Random seed. The same seed, start time, scenario and commands give the same data. Env `ARL_SEED`. |
| `speedup_multiplier` | float, × | `1.0` | Simulation speed relative to real time, up to the service's `max_speedup`. Env `ARL_SPEEDUP`. |
| `control_enabled` | boolean | `true` | Whether the service accepts commands. When `false` every command is rejected (`controls_disabled`). Env `ARL_CONTROL_ENABLED`. |
| `start_time` | `"now"` or RFC 3339 | `"now"` | Simulated time at start-up. Env `ARL_START_TIME`. |
| `scenario` | string, path | `""` | Scenario file to load, empty for none. Env `ARL_SCENARIO` (compose file). |
| `mqtt.address` | string | `"mosquitto"` | MQTT broker host. Env `MQTT_BROKER_ADDRESS`. |
| `mqtt.port` | integer | `1883` | MQTT broker port. Env `MQTT_BROKER_PORT`. |
| `mqtt.user` | string | `""` | Broker username, empty for anonymous. Env `MQTT_BROKER_USER`. |
| `mqtt.password` | string | `""` | Broker password. Env `MQTT_BROKER_PASSWORD`. |
| `mqtt.queue_capacity` | integer, messages | `200000` | Outgoing messages the service buffers while the broker is slow or unreachable. |
| `api.address` | string | `"api"` | ARL API host. Env `ARL_API_ADDRESS`. |
| `api.port` | integer | `8080` | ARL API port. Env `ARL_API_PORT`. |
| `api.require_researcher` | boolean | `true` | Whether commands must carry a researcher code the API accepts for this service. |
| `api.researcher_cache_ttl_s` | integer, s | `60` | How long a researcher check is remembered. |
| `api.alert_limits_refresh_s` | integer, s | `300` | How often the alert limits are re-read from the API. |

| Key | Type / unit | Default | Description |
|---|---|---|---|
| `health.port` | integer | `9100` | Port of the service's `/healthz` endpoint (used by `healthcheck`). Env `HEALTH_PORT`. |

### 5.0.2 OPG keys (`[opg]`)

| Key | Type / unit | Default | Description |
|---|---|---|---|
| `frequency_hz` | float, Hz | `0.0167` | Reporting rate of the `power` streams (whole minutes; 0.0167 Hz is every minute). |
| `solar_kwp` | float, kW | `1200` | Peak DC power of the solar array. |
| `inverter_kw` | float, kW | `1000` | Maximum AC output of the inverter. |
| `battery_kwh` | float, kWh | `4000` | Capacity of the battery bank. |
| `battery_max_kw` | float, kW | `800` | Maximum battery charge or discharge power. |
| `generators` | integer | `2` | Number of diesel generators. |
| `generator_kw` | float, kW | `250` | Rating of each generator. |
| `fuel_tank_l` | float, L | `20000` | Fuel tank size. |
| `refuel_every_days` | integer, days | `7` | Days between fuel deliveries. |
| `initial_soc_pct` | float, % | `70` | Battery state of charge at start-up. |
| `initial_reserve_pct` | float, %, 10 to 80 | `20` | Battery reserve at start-up. |
| `initial_fuel_pct` | float, % | `80` | Fuel level at start-up. |
| `initial_soiling` | float, ratio 0 to 1 | `0.97` | Cleanliness of the solar panels at start-up (1 is clean). |
| `array_lat` | float, degrees | `-31.2297564` | Latitude of the solar array (for the weather model). |
| `array_lon` | float, degrees | `21.6889263` | Longitude of the solar array. |
| `max_speedup` | float, × | `500` | Highest `speedup_multiplier` the service accepts. |

# 6 Scenario events

Events with `service = "opg"` (format in *Running the ARL*). Weather (`service = "env"`) also affects the OPG: storms and dust reduce solar output, and heat and cold change the load.

| Fault | Entity | Params (default) | Effect |
|---|---|---|---|
| `inverter_fault` | `solar` | none | The inverter fails: no solar output (`inverter_status fault`). |
| `string_failure` | `solar` | none | Part of the array goes down: lower solar output (Solar Underperformance). |
| `battery_cell_fault` | `batteries` | none | A cell fault heats the bank and costs some battery health (Battery Temperature). |
| `generator_fail_start` | `diesel` | none | The next generator start fails (Generator Fault). |
| `generator_trip` | `diesel` | none | Running generators fail (Generator Fault). |
| `fuel_leak` | `diesel` | none | The fuel level drops steadily until the next refuel (Low Fuel). |

| Fault | Entity | Params (default) | Effect |
|---|---|---|---|
| `set_state` | any | `soc_pct`, `fuel_pct` | Sets the battery state of charge and/or the fuel level directly. |
| `soiling` | `solar` | `ratio` | Sets the panels' cleanliness; rain cleans them again. |
| `repair` | `solar`, `batteries` or `diesel` | none | Ends the faults on that entity. |

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**AI Declaration**: Claude Code (Opus 5.5) - generated from the arl-sim source, reviewed